

ABHISHEK REDDY MALREDDY

+1 479-283-7033 | www.abhishekreddy.one | amalredd@andrew.cmu.edu

EDUCATION

CARNEGIE MELLON UNIVERSITY

GPA: 4.0/4.0

Pittsburgh, PA

Master of Science in Artificial Intelligence Engineering - Materials Science and Engineering

Dec 2025

Selected Coursework: Computer Vision, Systems and Tool Chains for AI Engineers, Machine Learning and AI for Engineers, Learning for 3D Vision, Introduction to Deep Learning, Methods of Computational Materials Science.

NATIONAL INSTITUTE OF TECHNOLOGY CALICUT

Calicut, India

Bachelor of Technology in Mechanical Engineering

May 2022

SKILLS

Programming: Python, C, C++, SQL, Git.

Machine Learning: PyTorch, PyTorch3D, TensorFlow, Keras, Pyspark, Transformers, Reinforcement Learning.

Software: AWS, GCP, Apache Spark, Docker, Eclipse SUMO, Data Modeling, LaTeX

ACADEMIC PROJECTS

TrackAnything: Multi-modal Promptable Tracking in 3D (CMU)

Jan 2025 – Apr 2025

- Developing an ML model to generate 3D bounding boxes from text prompts (e.g., "walking between parked vehicles") and video, enabling scenario mining in autonomous driving using the Argoverse 2 dataset.
- Integrated LiDAR, map, and vehicle pose data using a transformer-based architecture inspired by PromptTrack.
- Enhanced scenario mining capabilities for self-driving applications by improving object tracking accuracy in urban environments.

Real-time Pedestrian Detection and Safety Alert System (CMU)

Jan 2025 – Apr 2025

- Designed a real-time pedestrian detection and safety alert system using NVIDIA Jetson Nano, with YOLO & MobileNet SSD models trained on CityPersons and Caltech datasets.
- Optimized edge AI inference with quantization, pruning, and TensorRT, leveraging DeepStream for urban traffic safety.
- Developing an intelligent safety alert mechanism that triggers audio-visual alerts when pedestrians enter the road during restricted signals and flashes warning signals for speeding vehicles at red lights in real-time.

FIFA Player Analysis and Performance Prediction (CMU)

Sep 2024 – Dec 2024

- Unified multi-year FIFA player data into a PostgreSQL database, enabling scalable analytics with PySpark.
- Built ML models (GBT, RF, MLP), achieving a 30% RMSE reduction and improving player performance predictions.

PROFESSIONAL EXPERIENCE

MOBILITY RESEARCH (IIIT-H Affiliated)

Hyderabad, India

Research Assistant **Advisor: Dr. Girish Varma**

May 2023 – Jun 2024

- IDD-AW : Robust Semantic Segmentation for Autonomous Driving in Adverse Conditions:**
 - Developed the large-scale IDD-AW dataset (5000 RGB-NIR images) for semantic understanding of Indian driving scenes under adverse weather (rain, fog, snow, lowlight).
 - Introduced the novel "Safe mIoU" metric to enhance safety evaluation of segmentation models by penalizing critical misclassifications overlooked by traditional mIoU.
- Reinforcement Learning for Traffic Optimization**
 - Simulated congested traffic scenarios using SUMO software, implementing SB3-based reinforcement learning models to optimize lane selection, resulting in reduced traversal times for emergency vehicles.
 - Integrated lane-level dynamics and vehicle-to-vehicle communication to train models that outperformed human strategies, enhancing traffic flow in diverse congested conditions.
- Published "IdD-AW: A Benchmark for Safe and Robust Segmentation of Drive Scenes in Unstructured Traffic and Adverse Weather"** in the Proceedings of the IEEE/CVF Winter Conference on Applications of Computer Vision (WACV), Jan. 2024.

GENPACT INDIA

Hyderabad, India

Management Trainee

Aug 2022 – Apr 2023

- Optimized incident database workflows, improving SLA compliance and resolution speed.
- Collaborated with cross-functional teams to analyze incident trends and streamline workflows, thereby reducing ticket resolution time and boosting overall service quality.

PEBBLE DLT

Calicut, India

Developer

May 2020 – Oct 2020

- Built and simulated a decentralized Pebble DLT transaction management system using unique public key identifiers, enabling timestamp propagation and node categorization.
- Developed a configurable simulation framework for real-time transaction processing with dynamic queuing.